

Hybrid Approach to Train Delay Prediction: An Integration of Analytical Model and Deep Learning Techniques

Xingtang Wu , Wenbo Lian , Min Zhou , *Member, IEEE*, Hairong Dong , *Senior Member, IEEE*, and Fang Fang , *Senior Member, IEEE*

Abstract—High-speed rail (HSR) operation has the distinctive features of rapid speed and high density. When external environment fluctuations or system malfunctions interfere with the normal operation of certain trains, it may lead to train delays or even cascading delays, substantially compromising the normal functioning of the HSR system and adversely affecting passengers' travelling experience. To tackle this, this article proposes a hybrid train delay prediction model, which can effectively aid dispatchers in making optimized dispatching strategies and help passengers in replanning their trips. Taking into account of analytical mode, we identify the factors that influence train running and divide them into three categories. Then, a transformer-based network structure is constructed for predicting the delay time of the target train at the target station. Moreover, a loss function is designed in according to the root mean square error (RMSE) and the delay variation rules. Subsequently, numerous experiments are conducted based on the actual operation data of Chinese HSR subnetwork, and the results illustrate that our proposed model outperforms the baseline model substantially, with the RMSE and the mean absolute error (MAE) improved by 18.39% and 12.16% at most respectively, thereby verifying the superiority of our proposed model.

Index Terms—Deep learning, delay prediction, high-speed rail (HSR), transformer.

I. INTRODUCTION

HIGH-SPEED rail (HSR), serving as the foundation of a comprehensive national transport system, holds a great significance to the growth of the nation's economy and the

Manuscript received 23 February 2024; revised 12 May 2024; accepted 25 July 2024. Date of publication 23 August 2024; date of current version 14 February 2025. This work was supported by the National Natural Science Foundation of China under Grant 61925302 and Grant 62103033. (Corresponding authors: Hairong Dong; Min Zhou.)

Xingtang Wu and Fang Fang are with the School of Control and Computer Engineering, North China Electric Power University, Beijing 100096, China (e-mail: xtangwu@ncepu.edu.cn; ffang@ncepu.edu.cn).

Wenbo Lian and Min Zhou are with the School of Automation and Intelligence, Beijing Jiaotong University, Beijing 100044, China (e-mail: 21111067@bjtu.edu.cn; minzhou@bjtu.edu.cn).

Hairong Dong is with the College of Electrical Engineering and Automation, Shandong University of Science and Technology, Qingdao 266590, China, and also with the School of Automation and Intelligence, Beijing Jiaotong University, Beijing 100044, China (e-mail: hrdong@bjtu.edu.cn).

Digital Object Identifier 10.1109/TIE.2024.3440505

connection of cities [1]. In China, the HSR network spans over 40 000 km, making it the world's largest and most extensive HSR network, comprising eight vertical and eight horizontal arterial lines.

Under such a large-scale networked operation pattern, disturbances in any part of the system may cause cascading effects and the probability of emergencies increases. Additionally, train operations can be disrupted by both external and internal issues, such as equipment failure or inclement weather, leading to primary delays. These primary delays may cause delays in subsequent trains due to track constraints, leading to secondary delays throughout the current line or the entire HSR network.

In actual operations, emergencies can lead to multiple train postponements and potentially have significant negative societal impacts. For the existing centralized train control (CTC) system, dispatchers are responsible for one HSR line or just a part of a line. In case of emergencies, there is no delay propagation system to obtain the spatiotemporal data of delays. Consequently, dispatchers must anticipate delay changes within the sections they manage. Therefore, forecasting delay propagation on a global scale remains challenging.

Due to the track constraints of the train route, delays will not only be propagated along the current running lines, disrupting the normal operation of subsequent trains, but they may even cause cascading delays across the entire railway network. Factors such as train headway time, section operation time, and the number of tracks in the station all influence the propagation of train delays. Accurately predicting the spatiotemporal range of delay propagation during emergencies is of great importance in helping dispatchers come up with effective train rescheduling strategies and passenger reorganization plans. To this end, research has been conducted to gain insights into the propagation of delays based on the train operation constraints and general rescheduling strategies. Mazzarello et al. [2] studied the delay propagation theory under intensive operation frequency. They analyzed the relationship between train density and delays in detail, and comprehensively considered the train running process and the characteristics of infrastructure to predict the train delay time; Goverde et al. [3], [4] systematically defined the primary delay and the associated secondary delay, and then proposed a max-plus algebraic theory-based delay propagation model according to the train timetable to predict train

delays. The influence degree of various kinds of issues on train delays was also studied by the maximum spectrum analysis to evaluate the robustness of train schedules and quantify the model performance; Burdett et al. [5] analyzed the cause of train delays and proposed a time-event diagram-based model to derive the primary delays, then the buffer time was used to prohibit the delay propagation; Wei et al. [6] formulated a switched max-plus algebra-based model that was instrumental in quantifying delay propagation throughout a HSR network, with an additional focus on assessing the dynamic influences of severe weather on train operations through the development of multiple system matrices. Wu et al. [7] meticulously examined the phenomena of delay propagation within extensive HSR networks during various types of emergencies, and, in response, pioneered an innovative breadth-first search algorithm within a two-layer network framework to accurately calculate secondary delays.

The aforementioned analytical method essentially takes into account operational constraints, as well as a particular type of rescheduling strategy (e.g., retiming to make the most of buffer time). This approach serves to characterize the trend of train states under emergency conditions to a certain extent. However, in real-world scenarios, train dispatchers employ a variety of strategies to mitigate the effects of delays caused by unforeseen events. These strategies include, but are not limited to, retiming, rerouting, reordering, and even holding trains. Each strategy varies in implementation timing adjustments can differ in magnitude, and order adjustments can vary in order. Incorporating the diverse and specific adjustment measures that dispatchers take during emergencies is essential for understanding the substantial variability introduced in actual train schedules. Different rescheduling strategies, such as adjusting schedules by varying amounts, changing the sequence of train operations, or altering routes, inevitably lead to significant variations in the train timetables. These variations are critical to acknowledge as they directly affect the efficiency and timeliness of train operations. The diversity and complexity of these strategies are influenced by numerous factors, including the degree of delay caused by the incident and the dispatcher's personal experience and habits. This variety makes it challenging to encapsulate the full range of adaptive responses in a simple model. Traditional modeling methods might not fully capture the variability and uncertainty inherent in the multiple adjustment strategies used by different dispatchers. To address these limitations, the data-driven approach leverages extensive historical data to mine the dynamics of train operations, incorporating varied dispatcher strategies already recorded in past scenarios. The advantage of a data-driven approach is its ability to reflect a comprehensive spectrum of dispatching styles, thereby aligning the model outputs more closely with actual operational evolutions.

Kecman et al. [8] proposed a Markov process-based model to reveal the dynamics of delays. Specifically, the dynamic process of train delays' temporal and spatial variation was attributed to a random process. Then the prediction accuracy and the dynamic probability distribution of all events were analyzed, and the reliability of prediction results was improved by continuously fitting the dynamic characteristics of delays; Yang et al. [9]

conducted an examination of delay origins, classifications, frequency, affected train counts, and the spatiotemporal distribution of these delays, utilizing historical operational data, and employed the Kolmogorov-Smirnov test method to assess the goodness of fit among distinct train delay distribution models, aiding in the identification of potential delays; Corman et al. [10] developed a train delay prediction model using Bayesian networks, which facilitated understanding the interrelationships among random variables and modeling train interdependencies; Lessan et al. [11] proposed a Bayesian network-based train delay prediction model to decipher the complexities and dependencies in train operations. They calculated the evolution of delay propagation across the railway network using a time event graph. Statistical methods provide a foundational approach to understanding delay propagation through the analysis of historical data. However, these methods may struggle in adapting to and interpreting complex data interrelations, which limits their effectiveness in accurately predicting train delays in dynamic and evolving operational environments.

The deep learning method, as a vital part of the artificial intelligence [12], [13], [14], is developed to solve the problem of reasoning [15], decision making [16], etc., and is successfully applied in a vast of fields such as autonomous driving [17], traffic prediction [18], Internet of Things [19], and biomedicine [20].

Deep learning approaches have shown potential advantages over traditional statistical data-driven methods in handling the intricacies of train delay propagation. These advantages include enhanced capability for processing complex interrelations within large datasets, which is critical for effective prediction in this domain [21], [22], [23], [24], [25]. Wu et al. [22] proposed rule-driven automation model based on random forest, including delay state labeling, link elasticity, and station elasticity index. The model can significantly improve the generalization of multivariate multistep prediction models for train delay prediction under different scenarios; Zhou et al. [23] proposed a transfer learning and ensemble learning of stochastic vector functional link networks to predict train delay propagation. The propagation range and propagation chain of train delays can be obtained through iterative classification and prediction algorithms; Huang et al. [24] comprehensively considered the impact of spatio-temporal features, time-series features, nontime series features, and the weather on train delays and proposed a deep learning model consisting of a fully connected neural network (FCN) network and two long short term memory (LSTM) networks to predict delays. Chen et al. [25] proposed a delay prediction model based on a multimodal deep learning framework. The model comprehensively considered the spatio-temporal characteristics and time series characteristics of train operation and used 3D convolutional neural networks (3D-CNN) to extract the spatiotemporal data characteristics of the train operation. Finally, the accuracy of the model was verified by the actual operation data of the HSR network.

The current work on delay prediction based on deep learning has achieved relatively satisfactory results. However, it remains a challenge to improve the prediction performance due to the

complexity of the HSR system, which is composed of an immense number of subsystems. The following challenges persist.

- 1) Complex data environment in HSR systems: the intricate complexity of HSR systems and the diverse nature of operational data, encompassing time-series, spatial, and static elements, present significant challenges. These complexities hinder the capability of purely data-driven models to effectively identify and utilize relevant features for accurate delay prediction.
- 2) Limitations of purely data-driven models and discrepancies in predictions: while data-driven models are useful, they often fall short in complex operational contexts, particularly when actual operational rules are factored in. Furthermore, these models sometimes exhibit significant discrepancies between predicted results and actual outcomes, undermining their reliability. Enhancing the robustness and accuracy of these models is therefore a critical requirement.

To address these challenges, our article proposes the integration of model-based techniques with deep learning. This hybrid approach aims to capitalize on the strengths of both methodologies: the model-based techniques' adept handling of complex and varied data, and the deep learning's advanced pattern recognition and predictive capabilities. Such integration is especially aimed at reducing the prediction discrepancies and enhancing the overall reliability and precision of the predictive models. The main contributions of this article are as follows.

- 1) For the various influencing factors impacting the train delay time, we have extracted all the factors according to the analytical model. Subsequently, these influencing factors are classified into three distinct categories: static factors, time-series factors, and spatio-temporal factors, depending on the mechanism of each factor to the train operation, as well as the characteristics of different factors.
- 2) A hybrid a deep learning framework is designed that leverages a FCN and two transformers to extract the nonlinear relationship between different factors and prediction targets for the three distinct feature categories. This framework takes full advantage of the transformer's powerful sequence processing capacity to extract the status of the target train in the presequence stations and the effect of the running state of the presequence trains at the destination station to achieve accurate forecasting of the train delay time. Moreover, we have designed a loss function that not only builds upon the root mean square error (RMSE) but also incorporates the principles of the train delay propagation mechanistic model. This approach effectively constrains the predicted delay outputs within a theoretical range, thereby preventing excessive deviations in the predicted values. By aligning our loss function closely with the underlying mechanics of train delay propagation, we ensure that our model's predictions remain within realistic and theoretically sound boundaries, further enhancing the model's performance.
- 3) The actual operational data of a subnetwork of Chinese HSR was leveraged to train and validate the proposed

method. Compared to the baseline models, the proposed model was able to significantly enhance the prediction performance, with a marked improvement in RMSE and the mean absolute error (MAE) of 18.39% and 12.16% at most, respectively. And the prediction accuracy achieve 88% when the prediction error is less than 1 min, indicating the superior performance of the model.

The remainder of this article is organized as follows: Section II introduces the feature extraction model and data processing rules. Section III details the proposed model. Section IV applies the model to real-world data to verify its effectiveness. Finally, Section V concludes this work.

II. PRELIMINARY

A. Features Extraction According to the Max-Plus Based Delay Prediction Model

Train operations are influenced by many factors, including track constraints, signal constraints, station capacity constraints, interlocking rules, etc. The arrival time of a train at the next station is determined by both the train's current departure time and the arrival time of preceding trains at that station. Considering all the operation rules and retiming strategies, a max-plus algebra based delay propagation model was proposed as follows [6], [7]:

$$T_r = (C \otimes T_r) \oplus T_s \oplus T_d \quad (1)$$

where T_r denotes the matrix representing actual operational times for each train at every station, reflecting the real-world circumstances of train arrival, dwell, and departure times, T_s embodies the scheduled operation time matrix for each train across all stations, encapsulating the intended timetable that trains are anticipated to adhere to under ideal operational conditions, T_d stands for the delay time encountered by each train at every station, a resultant of various potential factors such as operational complications, unfavorable weather, technical issues, among others, and C is indicative of the system constraints matrix encapsulating all operation-related restrictions. In the data-driven prediction model, our goal is to predict the arrival time of the train of interest at a specific station. The constraints related to the arrival time are as follows:

$$t_{a,i}^{r,k} = (t_{d,i-1}^{r,k} \oplus t_{i-1,i}^{\min}) \otimes t_{a,i}^{s,k} \quad (2)$$

$$t_{a,i}^{r,k} = (t_{a,i}^{r,k-1} \oplus h_{k-1,i}^{a,a,\min}) \otimes t_{a,i}^{s,k} \quad (3)$$

where $t_{a,i}^{r,k}$ denotes the actual arrival time of train k in station i , $t_{d,i-1}^{r,k}$ denotes the actual departure time of train k in station $i-1$, $t_{i-1,i}^{\min}$ is the minimum allowed running time in section $[i-1, i]$, $h_{k-1,i}^{a,a,\min}$ is the minimum allowed headway time between adjacent trains. It can be derived from the constraints that the arrival time of train k in station i is directly dependent on the departure of train k in station $i-1$, the arrival time of train $k-1$ in station i , the headway time between two adjacent trains, the location of the station, the label of the train.

Considering these factors, we divide all features into three categories based on their characteristics and association with the trains: spatial-temporal features (f_1); time-series features

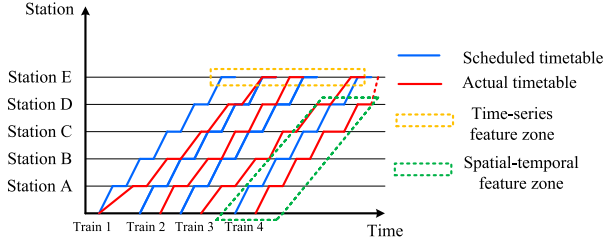


Fig. 1. Illustration of different features.

(f_2); and static features (f_3). f_1 indicates the time-related features and spatial-related features of the target train before the target station. For the target train, both the location and the operation time are variables and the number of considered stations is denoted as N_S . f_2 includes the time-related features of the former trains at the target station. The number of preceding trains is denoted as N_T . For those trains, the location is determined, while the arrival time of the succeeding train is dependent on the preceding train's arrival time. f_3 involves infrastructure-related, operation-related, and other features. Those features keep constant for the prediction target. It's worth highlighting that while in the aforementioned model the arrival time of a train directly correlates with its departure time from the previous station and the arrival time of the preceding train at the current station, the actual propagation of delay is a chain reaction in practical scenarios. To extrapolate the actual arrival time of a train at a station comprehensively from historical operational data, data from previous stations and preceding trains are also utilized. In subsequent experiments, the number of additional trains and stations involved is determined through experimental analysis. An illustration is shown in Fig. 1. The goal is to predict the arrival delay of train 4 in station E. f_1 indicate the spatial-temporal features of train 4 in station A, station B, station C, and station D, f_2 indicate the time-related features of train 1, train 2, and train 3 in station E; The detail of each type of features are listed as follows.

f_1 : scheduled arrival time, actual arrival time, scheduled departure time, actual departure time, and minimum section running time;

f_2 : scheduled arrival time, actual arrival time, scheduled headway time, actual headway time, and minimum headway time;

f_3 : train index, train type, direction, scheduled arrival time of the target train in the target station, scheduled headway time between the target train and the former train in the target station, target station name, and target station order.

B. Data Preprocessing

This study addresses the research problem of predicting future arrival delays at stations for trains that have already experienced delays. Consequently, scenarios where a train experiences no delay at station i , but incurs a delay at station $i + 1$ are not within the purview of this model. Such situations typically arise due to unforeseen incidents. Without incorporating information related to these unexpected events, the model

cannot capture delays caused by unknown incidents. Therefore, during data processing, any data depicting scenarios where a train incurs no delay at station i but experiences a delay at station $i + 1$ should be discarded.

Furthermore, considering that trains predominantly operate on schedule and the focus of our task is on circumstances where delays have already occurred, data sequences without delays are excluded (accounts for 90.22% of the total data set). This measure aims to enhance the efficiency of model training by centering the learning process around instances of delay.

Then a deep learning network named F-DTransformer that contains two transformers and a FCN is designed to predict the arrival delay of the target train in the target station. Specifically, one transformer is used to deal with f_1 , another is used to deal with the f_2 , and the FCN is used to deal with the f_3 . The detail of the model is introduced in the follows.

III. F-DTRANSFORMER FOR DELAY TIME PREDICTION

A. Transformer

The transformer is first proposed by [26] in 2017 for sequence prediction, and is widely used in machine translation, summary generation, computer vision, etc. The transformer adopts the Seq2Seq architecture and contains a multilayer encoder-decoder structure composed of stacked self-attention and fully connected layers. The encoder is responsible for mapping the input data into hidden layers, while the decoder is responsible for parsing the features of the hidden layer into output features. The detail steps of the transformer will be introduced as follows: initially, the input sequence $X = [x_1, x_2, \dots, x_N]$ is embedded into a $N \times d$ matrix X_E , where N symbolizes the input sequence length, and d denotes the encoding length of each sequence element. Subsequently, the input sequence's position information is encoded into a $N \times d$ matrix P_E to achieve a vector representation encompassing order and relative position details.

A transformer may contains several encoder unit, and each encoder consists of a multihead attention layer and a feed forward FCN layer. The multihead attention unit consists of several self-attention units, and the self-attention generally adopts the query-key-value pattern to get the output of the attention. The input matrix X_{input} is linearly mapped into Q , K , and V with the following equations:

$$Q = W_q X_{\text{input}} \quad (4)$$

$$K = W_k X_{\text{input}} \quad (5)$$

$$V = W_v X_{\text{input}} \quad (6)$$

where W_q , W_k , W_v are the mapping parameter matrix. Then the output of the self-attention unit is calculated as follows:

$$Z = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (7)$$

where d_k is a scaling factor to avoid the problem of vanishing gradients. A multihead attention unit is composed of several parallel self-attention units, in which different W_q , W_k , and W_v are adopted to generate different Q , K , and V for the different

self-attention units. Then the output of each self-attention unit is concatenated and merged as follows:

$$Z_m = \text{Concat}(Z_1, Z_2, \dots, Z_n)W^o \quad (8)$$

where Z_n is the n th self-attention output, and matrix W^o is a parameter to be trained. Then an add & norm unit is implemented to map the output of the multihead attention layer. The add & norm is implemented as follows:

$$\hat{Z}_m = \text{LayerNorm}(X_{\text{input}} + Z_m) \quad (9)$$

where $X_{\text{input}} + Z_m$ is a residual connection, making the network only focus on the part of the current difference, and the LayerNorm action normalizes each vector data vertically, which can speed up the convergence of training. The feed-forward FCN network consists of two layers, with the first layer using the ReLU activation function. The formula is as follows:

$$X_E = \max(0, \hat{Z}_m W_1 + b_1)W_2 + b_2. \quad (10)$$

The dimension of X_E is the same as $\hat{Z}_m$. The input $X_{(N \times d)}$ is mapped into $X_{E(N \times d)}$ by an encoder unit. For subsequent encoders, its input is the output of the previous encoder.

In this article, our prediction target is the delay time of the train of interest at a specific station, resulting in the output being a single-dimensional vector. To accomplish this task, we exclusively utilize the encoder module. That is, the output of the Encoder is directly fed into a linear layer followed by a softmax layer.

B. The F-DTransformer

In order to develop a deep learning network structure that performs well with diverse data attributes, a hybrid model that is composed of two transformer networks and a FCN network is designed. One transformer is responsible for processing time-series features, another for spatial-temporal features, while the FCN handles static features. Specifically, the outputs of the three networks are directly concatenated without using an activation function layer. Then the concatenated vector is input to a FCN and an active function to get the prediction results. In this way, it is possible to consider the impact of the considered train's operation status in preorder stations, the previous trains' operation status in the target station, and infrastructure-related factors on the arrival time of the considered train at the target station. The transformer exhibits a powerful ability to process sequential data, while the FCN adeptly handles static features. This combination leverages the strengths of both networks, thereby enhancing the model's overall performance.

C. Loss Function

Under normal circumstances, delay time tends to gradually decrease along the direction of the train's operation, barring any unforeseen incidents. Exceptions arise only in rare cases where the delay time may exhibit a slight increase, particularly when the number of trains dwelling exceeds the maximum allowable limit, as demonstrated in Fig. 2. The observation reveals that

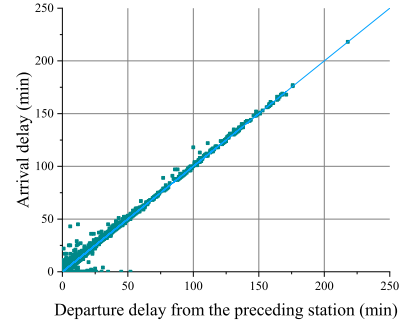


Fig. 2. Distribution of arrival delay and departure delay in former station.

the arrival delay predominantly aligns with or is less than the departure delay from the preceding station in most instances.

To incorporate these constraints into the prediction model, we propose a novel modification of the standard RMSE loss function, articulated as follows:

$$R_{\text{RMSE}} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\min(\max(0, y'_i), y_i^{\text{up}}) - y_i)^2} \quad (11)$$

$$\text{Penalty} = \frac{1}{n} \sum_{i=1}^n \max(0, y'_i - y_i^{\text{up}}) \quad (12)$$

$$L(y, y') = R_{\text{RMSE}} + \text{penalty_factor} \times \text{Penalty} \quad (13)$$

where y'_i is the predicted value, y_i is the label, and y_i^{up} is the upper limit that y'_i intended to hold, and penalty_factor is an parameter. y_i^{up} is defined as follows:

$$y_i^{\text{up}} = d_i + \aleph \quad (14)$$

where d_i is the departure delay time of this train in the former station, and $\aleph$ acts as a relaxation parameter to account for scenarios where the number of trains dwelling surpasses the prescribed maximum limit. The penalty term is designed to penalize predictions where the predicted delay at a station exceeds the delay at the previous station. Specifically, the penalty is applied when $y'_i > y_i^{\text{up}}$, where y_i^{up} represents the upper bound of the predicted delay. The penalty factor in the loss function plays a crucial role in balancing the importance of the RMSE and the penalty term. Its value is determined based on empirical testing to optimize the model's predictive accuracy while ensuring that the model does not overly penalize deviations that are within a reasonable range. Employing this loss function ensures that the predicted delay time exhibits a progressive decrease along the train's operational direction, thereby enhancing the robustness and precision of the model's predictions. Moreover, the Relu function is selected as the activation function in all experiments.

D. Performance Indicators

Two indicators, namely RMSE and MAE, are adopted to evaluate the performance of the model. RMSE and MAE are defined as follows:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{n=1}^N (y_i - y'_i)^2} \quad (15)$$

$$\text{MAE} = \frac{1}{N} \sum_{n=1}^N |y_i - y'_i| \quad (16)$$

where N is the size of the testing data.

E. Hyperparameter Optimization

Bayesian optimization is a probabilistic model-based approach for global optimization of costly-to-evaluate functions, including hyperparameters in deep learning. This method models the unknown function as a Gaussian process, then applies a utility function, like expected improvement, to balance exploration against exploitation. It intelligently selects the most promising hyperparameters to evaluate next by incorporating prior observations, thereby reducing the required number of function evaluations. This efficiency and effectiveness make Bayesian optimization a leading choice for hyperparameter optimization in deep learning models. The optuna is a flexible, lightweight, and versatile Python library for hyperparameter optimization, built to automate the process and improve the performance of machine learning models. Indeed, Optuna has effectively integrated Bayesian optimization into its framework, making it a practical choice for hyperparameter optimization in deep learning models. By utilizing the Bayesian optimization features of Optuna, we can streamline the process of finding the most effective hyperparameters for our model. This not only reduces the computational resources required but also enhances the overall performance of the deep learning model. In the subsequent sections, we will apply the Optuna framework, leveraging its Bayesian optimization capabilities, to optimize our model's hyperparameters.

F. Benchmarks

To evaluate the effectiveness of the proposed model, several benchmark models are utilized for comparison, which are briefly introduced as follows: (1) FCN (M_1): all the features are concatenated into a sequence, and the order of different input dimension will not affect the results. Then the delay time of the target train in the target station is set as the output. (2) RF(M_2): the input and output processes for the RF are similar to those of the FCN. (3) XGBoost (M_3 [27]): similar to the FCN, XGBoost utilized identical methods for input and output. (4) LSTM (M_4 [28]): concerning the delay prediction task, a duo of LSTM networks and a FCN were deployed. One LSTM was charged with handling f_1 , while the other managed f_2 . The FCN took care of f_3 . The final output of this model corresponded to the delay time for the target train at the destined station. (5) F-DTransformer (M_5): the RMSE is selected as loss function for those models. Also in order to show the effectiveness of modified loss function, a F-DTransformer with the modified loss function is denoted as M_6 . In the following, all the model are trained with the real operation data, and the flowchart of those delay prediction model is shown in Fig. 3.

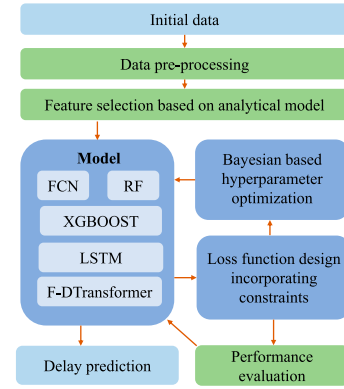


Fig. 3. Flowchart of the delay prediction model.

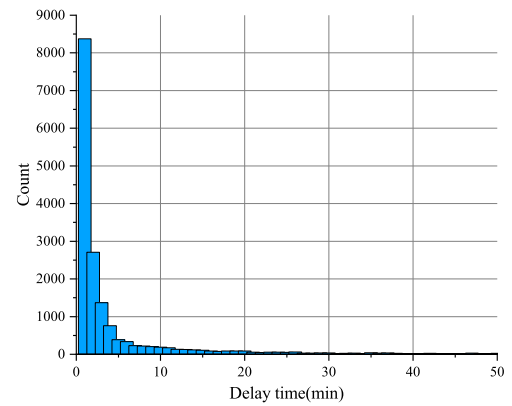


Fig. 4. Distribution of delayed records.

TABLE I
PERFORMANCE INDICATORS FOR DIFFERENT MODELS

	M_1	M_2	M_3	M_4	M_5	M_6
RMSE	3.6563	11.22	1.8756	1.9244	1.6963	1.5309
MAE	1.7243	3.3401	0.7723	0.7268	0.6657	0.6384

Note: Bold entries indicate the outcomes generated by the proposed model.

IV. EXPERIMENTS BASED ON THE CHINESE HSR OPERATION DATA

A. Dataset Description

In this article, the real operation data and the infrastructure data of the Chinese HSR within a Railway Administration are adopted to verify the performance of the proposed model. The data ranges from 3 June 2019 to 31 October 2019, a total of 1.98 billion records. After deleting the abnormal value through data cleaning, a total of 1.39 billion data remains. Among all the recorded data, there are 0.29 billion pieces of delayed data. The arrival delay distribution of all data is shown in Fig 4. It can be observed that as the delay time escalates, there's

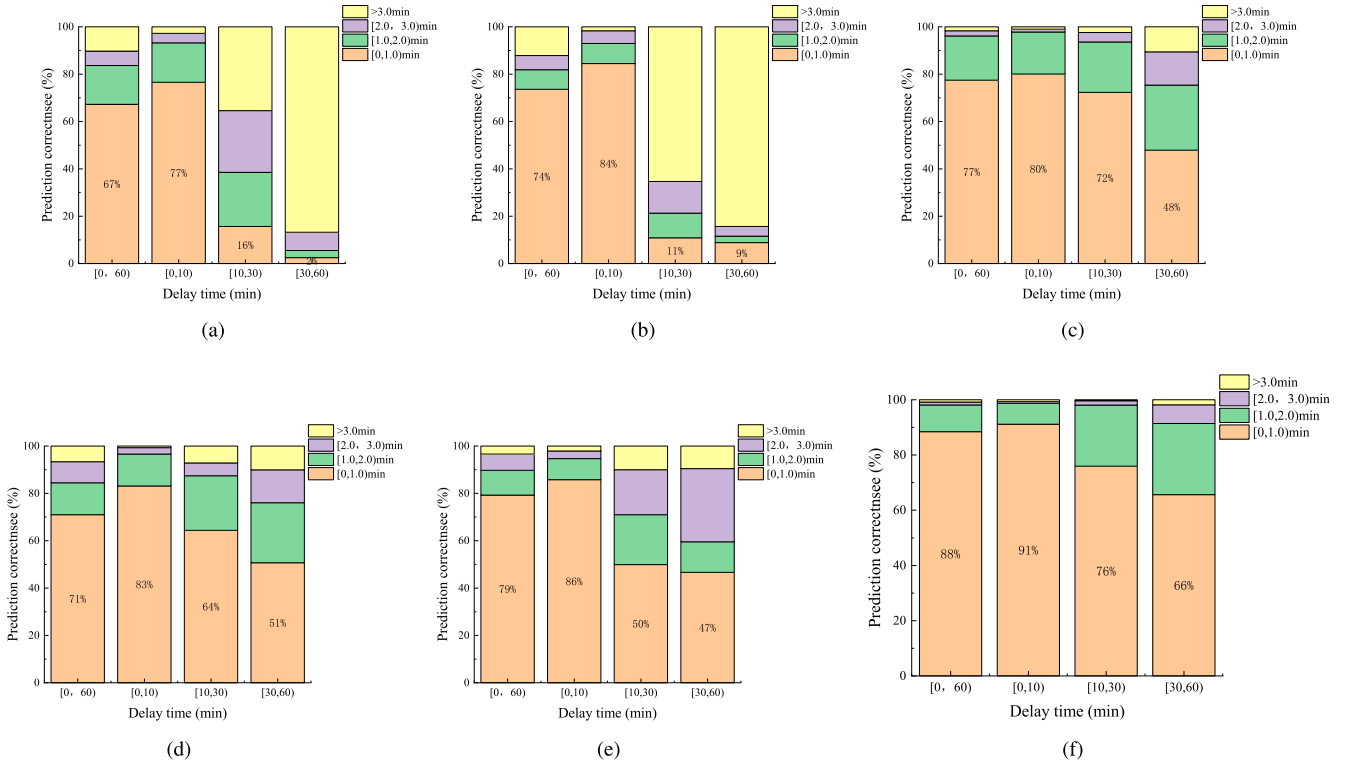


Fig. 5. Prediction error distribution for different models: (a) M_1 ; (b) M_2 ; (c) M_3 ; (d) M_4 ; (e) M_5 ; and (f) M_6 .

a gradual decline in the number of delayed samples. For instance, there are 16 306 records with a delay time of less than 30 min, while records with a delay time exceeding 60 min only amount to 346.

All the experiments are coded with Python 3.8, and the pytorch framework is adopted to construct the deep learning network. ALL experiments are implemented on a personal computer with 2.10GHz Intel Xeon Gold 6130 CPU, NVIDIA GeForce RTX 3090 GPU, 16GB RAM with Windows 10 system.

B. Performance Evaluation

First, all hyperparameters of each model are optimized via the Bayesian Optimization for fair comparison, e.g., the learning rate, batch size, dropout rate, number of attention heads, hidden dimensions, number of encoder unit, are the hyperparameters for the M_6 .

All the models are trained and verified with the same data set. The performance of each model in terms of RMSE and MAE are shown in Table I.

It is evident that the proposed model structure excels over the benchmark models in both RMSE and MAE metrics. Additionally, the implementation of the modified loss function results in a marked enhancement in the model's performance, specifically contributing to a 9.75% improvement in RMSE and a 4.10% enhancement in MAE. These substantial gains demonstrate that the proposed model is not only effective but also represents a

significant advancement in model performance, reflecting its capability to address complex challenges in the given context.

In order to further validate the efficacy of the proposed model, we also evaluate its prediction accuracy across different delay time levels. As shown in Fig. 5, the distribution of prediction errors within different delay time segments is presented.

From the observed results, it is clear that the model M_5 outperforms the benchmark models. The M_5 model demonstrates superior performance in terms of prediction accuracy when compared to other models. Specifically, it attains a prediction accuracy of 79% within the permitted error segment of [0,1) min, while the best of the benchmark models reaches 77%, and the worst is confined to only 67%. Additionally, the prediction accuracy of M_6 exhibits a substantial improvement, achieving an impressive 88% accuracy within the same error segment. Furthermore, when considering a delay time interval of [0,10) min, M_6 continues to outshine the competition, with prediction accuracy exceeding 91% when the prediction error is less than 1 min.

An observation arises when examining the relationship between the delay time interval and the prediction accuracy rate. As the delay time interval increases, there is a noticeable decline in prediction accuracy. This is due to the sparse nature of the dataset for large delays, which negatively impacts the performance of each model. The decline is particularly steep for models M_1 and M_2 , where the accuracy drops dramatically in the time interval of [30,60) min. Conversely, models M_5 and M_6 maintain satisfactory results even within this challenging interval, e.g., the prediction error exceeding 3 min only accounts

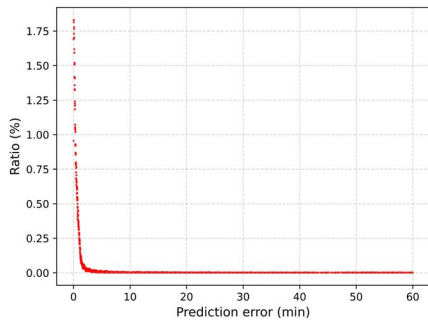


Fig. 6. Prediction error distribution for F-Dtransformer.



Fig. 7. Delay prediction cases.

for 1.84% when the delay time interval is [30,60) min. This is because that the transformer architecture in our models is crucial for effectively handling the complexities associated with larger delays, particularly in scenarios where data sparsity is a prevalent challenge. Additionally, the FCN component complements this by adeptly processing static features, a necessary aspect in delay prediction tasks. Furthermore, our refined loss function, designed to keep predicted values within a theoretical range, significantly boosts the models' predictive accuracy. This aspect is especially vital in addressing the variability and uncertainty often encountered in delay prediction. The combination of these advanced methodologies enables models $M5$ and $M6$ to maintain robust performance and deliver satisfactory results, even in challenging conditions characterized by larger delays and sparse data. Together, these improvements solidify the proposed models as robust and reliable solutions for the task at hand.

Fig. 6 also reveals a detailed distribution of the prediction error for the M_6 , indicating that, in the majority of cases, the prediction error is kept at a low level. Based on the experimental analysis above, it is evident that the proposed method in this article can effectively predict train delays under various conditions and demonstrates a significantly better performance than existing methods.

C. Prediction Cases

To demonstrate the efficacy of the proposed model, several prediction cases are presented in Fig. 7. A subsegment of the Beijing-Guangzhou HSR line is selected, and the timetable of train G807 from 4 June 2019 is utilized. The figure showcases the actual delay time of G807 and its predicted delay time at four stations using the proposed model. Even with a delay time approaching 20 min, the prediction error is consistently

under 1 min, ensuring its suitability for real-time operational requirements.

V. CONCLUSION

The HSR has seen significant growth, featuring rapid speed and frequent service, evolving into a networked operation. Any disruptions can lead to a chain reaction of delays at both line and network levels. Accurate and timely prediction of train delays can help dispatchers devise optimized strategies, effectively minimizing the negative impact of emergencies. To address this challenge, this article presents a hybrid train delay prediction model that combines both analytical and deep learning approaches. Specifically, train operation-influencing features are extracted from the analytical model and divided into three categories: time-series, spatial-temporal, and static features. A hybrid deep learning model named F-DTransformer is then developed, consisting of a FCN and two transformers. The FCN processes static features, while the transformers handle time-series and spatial-temporal features, respectively. A specialized loss function based on the RMSE and delay time variation rule is also formulated. Experiments on real Chinese HSR sub-network data show the proposed model substantially improves prediction performance, yielding an 18.39% improvement in RMSE and a 12.16% enhancement in MAE compared to the best benchmark models. Furthermore, the model achieves 88% prediction accuracy when the permitted error is under 1 min, highlighting its advantages.

REFERENCES

- [1] W. Wu, H. Song, H. Wang, and H. Dong, "Potential game based task offloading in the high-speed railway with reinforcement learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 11, pp. 12671–12685, Nov. 2023.
- [2] M. Mazzarello and E. Ottaviani, "A traffic management system for real-time traffic optimisation in railways," *Transp. Res. Part B Methodol.*, vol. 41, no. 2, pp. 246–274, 2007.
- [3] R. M. Goverde, "Railway timetable stability analysis using max-plus system theory," *Transp. Res. Part B Methodol.*, vol. 41, no. 2, pp. 179–201, 2007.
- [4] R. M. Goverde, "A delay propagation algorithm for large-scale railway traffic networks," *Transp. Res. Part C Emerg. Technol.*, vol. 18, no. 3, pp. 269–287, 2010.
- [5] R. Burdett and E. Kozan, "Determining operations affected by delay in predictive train timetables," *Comput. Oper. Res.*, vol. 41, pp. 150–166, 2014.
- [6] D. Wei, H. Liu, and Y. Qin, "Modeling cascade dynamics of railway networks under inclement weather," *Transp. Res. Part E Logistics Transp. Rev.*, vol. 80, pp. 95–122, Aug. 2015.
- [7] X. Wu, M. Yang, W. Lian, M. Zhou, H. Wang, and H. Dong, "Cascading delays for the high-speed rail network under different emergencies: A double layer network approach," *IEEE/CAA J. Automatica Sinica*, vol. 10, no. 10, pp. 2014–2025, Oct. 2022.
- [8] P. Kecman, F. Corman, and L. Meng, "Train delay evolution as a stochastic process," in *Proc. 6th Int. Conf. Railway Oper. Model. Anal. (RailTokyo)*. IVT, ETH Zurich; Orange Labs, 2015.
- [9] Y. Yang, P. Huang, Q. Peng, J. Li, and C. Wen, "Statistical delay distribution analysis on high-speed railway trains," *J. Modern Transp.*, vol. 27, no. 3, pp. 188–197, 2019.
- [10] F. Corman and P. Kecman, "Stochastic prediction of train delays in real-time using bayesian networks," *Transp. Res. Part C Emerg. Technol.*, vol. 95, pp. 599–615, Oct. 2018.
- [11] J. Lessan, L. Fu, and C. Wen, "A hybrid Bayesian network model for predicting delays in train operations," *Comput. Ind. Eng.*, vol. 127, pp. 1214–1222, Jan. 2019.

- [12] C. Bhatt, I. Kumar, V. Vijayakumar, K. U. Singh, and A. Kumar, "The state of the art of deep learning models in medical science and their challenges," *Multimedia Syst.*, vol. 27, no. 4, pp. 599–613, 2021.
- [13] Y. Pan, Z. Han, T. Wu, and Y. Lei, "Remaining useful life prediction of lubricating oil with small samples," *IEEE Trans. Ind. Electron.*, vol. 70, no. 7, pp. 7373–7381, Jul. 2023.
- [14] Y. Li, A. Zhang, J. Huang, and Z. Xu, "An approach based on transfer learning to lifetime degradation rate prediction of the dry-type transformer," *IEEE Trans. Ind. Electron.*, vol. 70, no. 2, pp. 1811–1819, Feb. 2023.
- [15] O. Li, H. Liu, C. Chen, and C. Rudin, "Deep learning for case-based reasoning through prototypes: A neural network that explains its predictions," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, no. 1, pp. 3530–3537, 2018.
- [16] S. J. Fong, G. Li, N. Dey, R. G. Crespo, and E. Herrera-Viedma, "Composite monte carlo decision making under high uncertainty of novel coronavirus epidemic using hybridized deep learning and fuzzy rule induction," *Appl. Soft Comput.*, vol. 93, 2020, Art. no. 106282.
- [17] K. Muhammad, A. Ullah, J. Lloret, J. Del Ser, and V. H. C. de Albuquerque, "Deep learning for safe autonomous driving: Current challenges and future directions," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4316–4336, Jul. 2021.
- [18] H. Yao, X. Tang, H. Wei, G. Zheng, and Z. Li, "Revisiting spatial-temporal similarity: A deep learning framework for traffic prediction," in *Proc. AAAI Conf. Artif. Intell.*, 2019, vol. 33, no. 1, pp. 5668–5675.
- [19] Z. Lv, L. Qiao, J. Li, and H. Song, "Deep-learning-enabled security issues in the Internet of Things," *IEEE Internet Things J.*, vol. 8, no. 12, pp. 9531–9538, Jun. 2021.
- [20] H. Bolhasani, M. Mohseni, and A. M. Rahmani, "Deep learning applications for iot in health care: A systematic review," *Inform. Med. Unlocked*, vol. 23, 2021, Art. no. 100550.
- [21] L. Oneto et al., "Train delay prediction systems: A big data analytics perspective," *Big data Res.*, vol. 11, pp. 54–64, Mar. 2018.
- [22] J. Wu et al., "A hybrid LSTM-CPS approach for long-term prediction of train delays in multivariate time series," *Future Transp.*, vol. 1, no. 3, pp. 765–776, 2021.
- [23] P. Zhou, L. Chen, X. Dai, B. Li, and T. Chai, "Intelligent prediction of train delay changes and propagation using RVFLNs with improved transfer learning and ensemble learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 12, pp. 7432–7444, Dec. 2021.
- [24] P. Huang, C. Wen, L. Fu, Q. Peng, and Y. Tang, "A deep learning approach for multi-attribute data: A study of train delay prediction in railway systems," *Inf. Sci.*, vol. 516, pp. 234–253, Apr. 2020.
- [25] S. Chen, X. Wu, M. Zhou, B. Yang, J. Lü, and H. Dong, "Train delay prediction based on a multimodal deep-learning method," in *Proc. China Automat. Congr. (CAC)*, Piscataway, NJ, USA: IEEE Press, 2021, pp. 3241–3246.
- [26] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, pp. 1–11, 2017.
- [27] R. Shi, X. Xu, J. Li, and Y. Li, "Prediction and analysis of train arrival delay based on XGBoost and Bayesian optimization," *Appl. Soft Comput.*, vol. 109, 2021, Art. no. 107538.
- [28] J. Luo, P. Huang, and Q. Peng, "A multi-output deep learning model based on Bayesian optimization for sequential train delays prediction," *Int. J. Rail Transp.*, vol. 11, no. 5, pp. 705–731, 2022.



Xingtang Wu received the Ph.D. degree in traffic information engineering and control from Beijing Jiaotong University, Beijing, China, in 2020.

From 2016 to 2017, he was a Research Assistant with The Hong Kong Polytechnic University, Hongkong, China. From 2021 to 2023, he was a Postdoctoral Research Fellow of Zhuoyue Program with the School of Automation Science and Electrical Engineering, Beihang University, Beijing, China. Currently, he is a Lecturer with the School of Control and Computer Engineering,

North China Electric Power University, Beijing, China. His research interests include railway network modeling and performance optimization, delay propagation, and artificial intelligence.



Wenbo Lian received the M.S. degree in traffic information signal and control from Lanzhou Jiaotong University, Lanzhou, China, in 2021. He is currently working toward the Ph.D. degree in traffic information engineering and control with the School of Automation and Intelligence, Beijing Jiaotong University, Beijing, China.

His research interests include railway disruption management, the modeling of complex network, and train delay propagation.



Min Zhou (Member, IEEE) received the Ph.D. degree in traffic information engineering and control from Beijing Jiaotong University, Beijing, China, in 2019.

Currently, he is an Associate Professor with the School of Automation and Intelligence, Beijing Jiaotong University. His research interests include crowd emergency management and evacuation, intelligent rescheduling for high-speed trains, and parallel intelligence.



Hairong Dong (Senior Member, IEEE) received the Ph.D. degree in general mechanics and basic mechanics from Peking University, Beijing, China, in 2002.

She was a Visiting Scholar with the University of Southampton, Southampton, U.K., in 2006, and The University of Hong Kong, Hong Kong, in 2008. She was also a Visiting Professor with the KTH Royal Institute of Technology, Stockholm, Sweden, in 2011. Currently, she is the Vice President with Shandong University of Science

and Technology, Qingdao, China, a Professor with the School of Electrical Engineering and Automation, Qingdao, China, and a Professor with Beijing Jiaotong University, Beijing, China. Her research interests include modeling and optimization of intelligent transportation systems, autonomous perception, collaborative control, and artificial intelligence.

Dr. Dong is a Co-Chair of the Technical Committee on Railroad Systems and Applications of the IEEE Intelligent Transportation Systems Society and the Fellow of the Chinese Association of Automation. She serves as an Associate Editor for IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II: EXPRESS BRIEFS, IEEE TRANSACTIONS ON INDUSTRIAL CYBER-PHYSICAL SYSTEMS, IEEE OPEN JOURNAL OF THE INDUSTRIAL ELECTRONICS SOCIETY, IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS, IEEE TRANSACTIONS ON INTELLIGENT VEHICLES, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II: EXPRESS BRIEFS, and IEEE INTELLIGENT TRANSPORTATION SYSTEMS MAGAZINE.



Fang Fang (Senior Member, IEEE) received the M.Sc. degree in control theory and engineering from the North China Electric Power University, Baoding, China, in 2001, and the Ph.D. degree in thermal power engineering from the North China Electric Power University, Beijing, China, in 2005.

He is a Professor of control science and engineering and the Vice President of North China Electric Power University. He is leading an Inter-governmental Cooperation Projects of National

Key Research and Development Programs Project and a National Natural Science Foundation Project of China. He has authored more than 60 highly level publications, and headed more than 25 research projects or industrial projects. His research interests include Internet of Things technologies in modern power systems, optimal configuration and operation of integrated energy systems, and intelligent power generation technologies.

Dr. Fang is a Council Member of the Chinese Association of Automation, and a Senior Member of the Chinese Society for Electrical Engineering and the Chinese Association for Artificial Intelligence.